Stefan's Law

Aim

To verify that a body at temperature T radiates heat at a rate proportional to the fourth power of T and calculate the constant of proportionality.

Theory

For a body at temperature T and ambient temperature T_a, the rate of heat exchange is given by the expression

$$\dot{Q} = \epsilon\sigma A(T^4 - T_a^4)$$

where σ is Stefan's constant, ϵ is the emissivity and A is the surface area of the body. In the experimental setup, a tungsten filament is connected to a battery and emits radiation. When the system archives thermal equilibrium, the electrical power consumed is equal to the rate of heat radiated. It is also given that the resistance of the filament depends on the temperature according to the given equation

$$R = R_0(1 + \alpha T + \beta T^2)$$

$$\beta T^2 + \alpha T + (1 - \frac{R}{R_0}) = 0$$

$$T = \frac{-\alpha + \sqrt{\alpha^2 - 4\beta(1 - \frac{R}{R_0})}}{2\beta}$$

$$T = \frac{-5.21 + \sqrt{27.1441 - 2.88(1 - \frac{R}{1.31})}}{14.4} \times 10^4$$

Analysis

The actual dependence of temperature is given by

$$\frac{dR}{R} = cdT$$

where c is a constant. Solving this equation gives

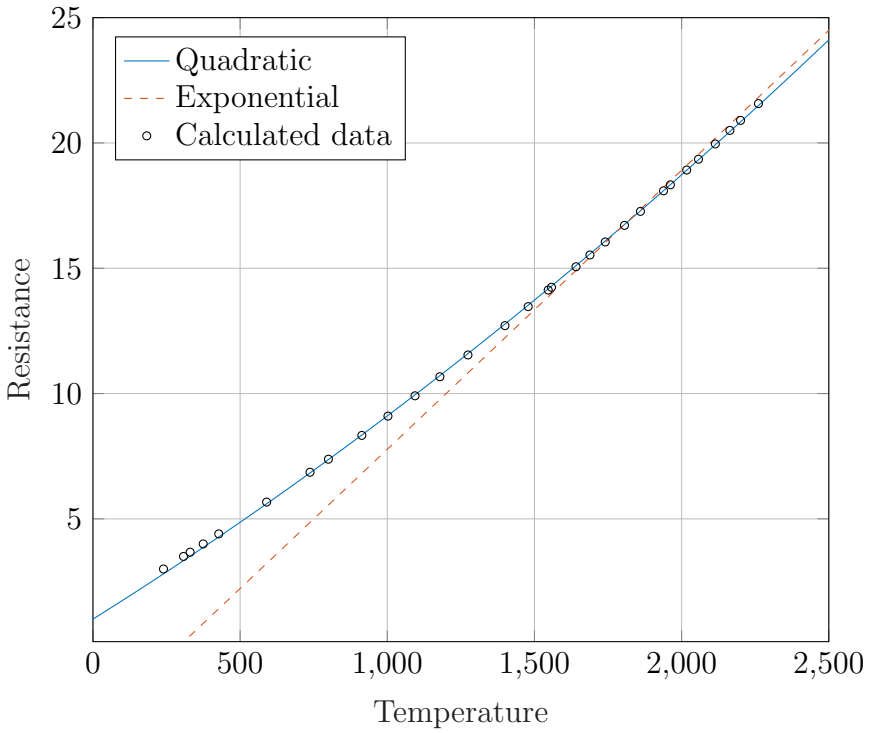
$$R = R_0 e^{c(T-T_0)}$$

The given relation of the dependence of temperature is the taylor series approximation This is a good approximation but has a few problems that are immediately apparent. First we see the fisrt few calculated values of resistance by the quadratic relation.

Voltage(V)	Current(I)	Resistance(R)	Temperature(T)
0.03	0.01	3.00	239.61
0.07	0.02	3.50	307.70
0.11	0.03	3.67	330.14
0.16	0.04	4.00	374.64
0.22	0.05	4.40	427.39
0.34	0.06	5.67	590.07
0.48	0.07	6.86	737.43
0.59	0.08	7.38	800.00

Table 1: Values of R and T for lower values

Notice that the first value of the calculated resistance is more than $50K$ below the ambient temperature ($296K$) which cannot be possible. To see what is going wrong, we look at the graph below.



In the graph,all data points of T calculated from resistance are ploted against R. The solid line represents the best fit quadratic curve which is the same as the one listed above. The

data points match up because this is the relation used to calculate the values of temperature. The dotted line shows the best fit exponential curve.

From the figure we see that the quadratic curve almost indistinguishable from the exponential near the higher values, however at lower values of y both curves begin to diverge. For the same value of y the value of x for the quadratic function is lower than the value of x for the exponential. This means that for lower values of resistance, the quadratic formula for temperature gives a lower than expected value which is most likely the reason that the first value of calculated temperature is lower than the ambient temperature

For this reason, I shall be excluding all values lower than the given value of $800K$ of glowing temperature.

The second question in the points to ponder asks how much error is in neglecting the T_a term in the heat exchange expression. It is safe to assume that the ambient temperature ranges between $300K$ to $400K$.(input source)

For this we see the following table

T_a	$T^4 - T_a^4$
0	4.10E+11
20	4.10E+11
40	4.10E+11
60	4.10E+11
80	4.10E+11
100	4.10E+11
120	4.09E+11
140	4.09E+11
160	4.09E+11
180	4.09E+11
200	4.08E+11
220	4.07E+11
240	4.06E+11
260	4.05E+11
280	4.03E+11
300	4.02E+11
320	3.99E+11
340	3.96E+11
360	3.93E+11
380	3.89E+11
400	3.84E+11

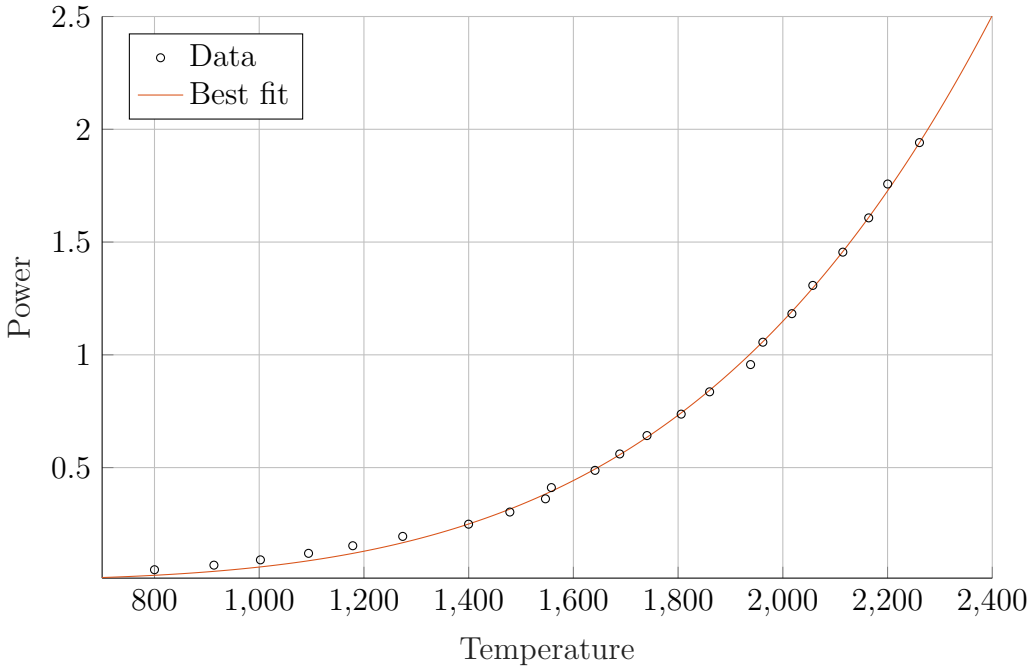
Table 2: Values are calculated for T=800K

From the table we can calculate the worst case error in neglecting the ambient temperature

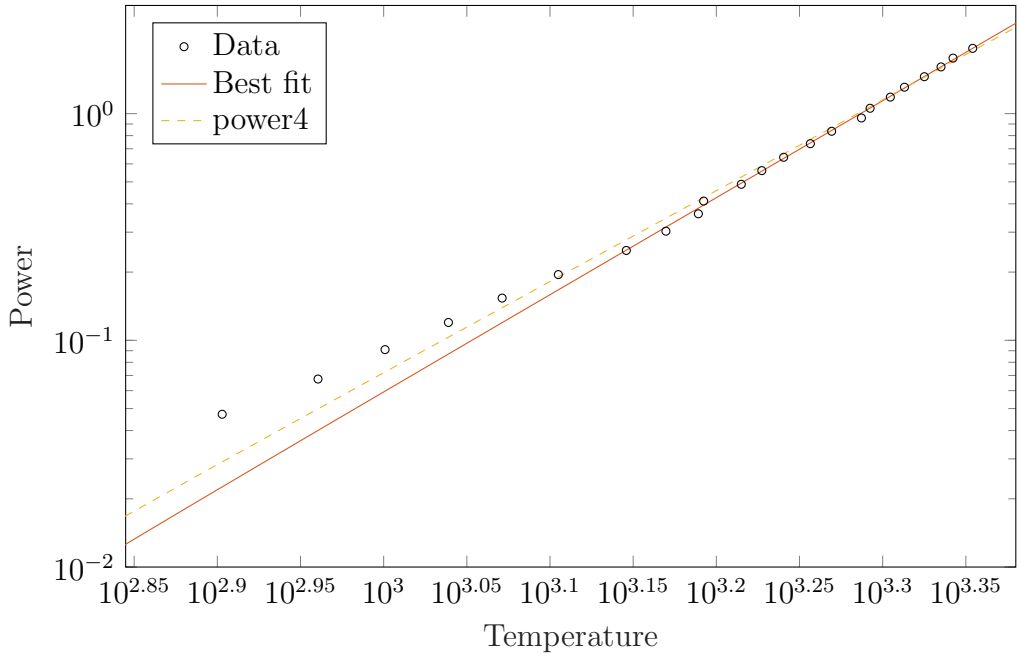
$$\frac{4.10 - 3.84}{3.84} = 6.77\%$$

As the temperature increases, due to the magnitude of T^4 the error becomes small.

I shall not be ignoring the ambient temperature completely, but considering it to be constant at arount $300K$.



Performing regression for the formula $y = c(x^a - 300^a)$ we get the value of a as 4.27 (with $R^2 = 0.9987$)



When fitting is performed with the formula $y = c(x^4 - 300^4)$ we get the value of c to be 7.24376×10^{-14}

Now, $c = \epsilon\sigma A$. The fillament can be thought of as a cylinder, thus the surface area is $2\pi rh$. The radii(r) and length(h) of the filament were the order of 10^{-3}(1mm) each. ϵ lies between 0 and one. This verifies that the order of magnitudes of calculated and given Stefan-Boltzmann constant are the same.

$$\frac{10^{-14}}{10^{-3} \times 10^{-3}} = 10^{-8}$$

There is not really much to comment about the value because we do not know the value of ϵ and A

Result

We have verified that heat radiated is proportional to the fourth power of temperature.